

2.6.38

AI25BTECH11027 - NAGA BHUVANA

Question:

If $\mathbf{a} = \hat{i} + \hat{j} + \hat{k}$ and $\mathbf{b} = \hat{j} - \hat{k}$, find a vector $\mathbf{c}$ such that $\mathbf{a} \times \mathbf{c} = \mathbf{b}$ and $\mathbf{a} \cdot \mathbf{c} = 3$

Solution:

Given that

$$\mathbf{a} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \quad (1)$$

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \quad (2)$$

By vector triple product

$$\mathbf{a} \times (\mathbf{a} \times \mathbf{c}) = \mathbf{a} \times \mathbf{b} \quad (3)$$

$$(\mathbf{a}^T \mathbf{c})\mathbf{a} - (\mathbf{a}^T \mathbf{a})\mathbf{c} = \mathbf{a} \times \mathbf{b} \quad (4)$$

As $(\mathbf{a}^T \mathbf{c}) = 3$ and $(\mathbf{a}^T \mathbf{a}) = 3$

$$3\mathbf{a} - 3\mathbf{c} = \mathbf{a} \times \mathbf{b} \quad (5)$$

$$3 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - 3\mathbf{c} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \quad (6)$$

$$3\mathbf{c} = \begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix} - \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \quad (7)$$

$$3\mathbf{c} = \begin{pmatrix} 5 \\ 2 \\ 2 \end{pmatrix} \quad (8)$$

$$\mathbf{c} = \begin{pmatrix} \frac{5}{3} \\ \frac{2}{3} \\ \frac{2}{3} \end{pmatrix} \quad (9)$$

$$\therefore \mathbf{c} = \frac{5}{3}\hat{i} + \frac{2}{3}\hat{j} + \frac{2}{3}\hat{k}$$

Vectors a, b, and c

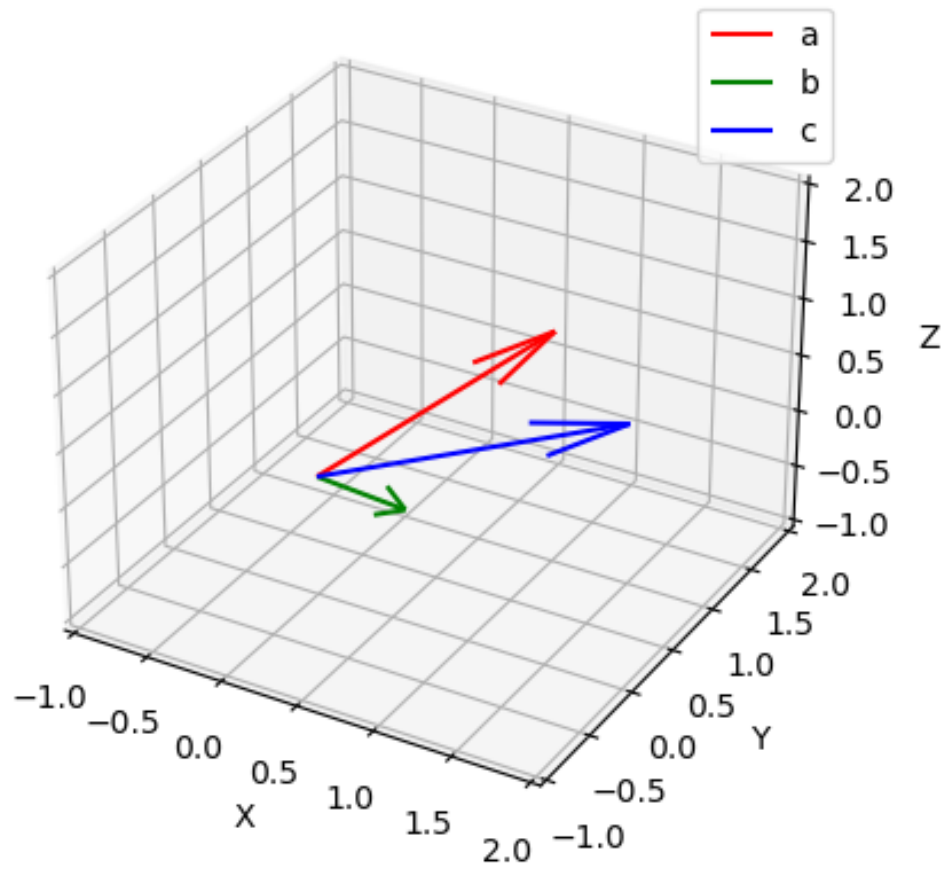


Fig. 1